

MYSQL PROJECT REPORT

RETAIL SALES DATA ANALYSIS

Submitted by

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Data Analytics Course

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ACKNOWLEDGEMENT

I would like to thank my trainer and Besant Technologies for providing guidance and the dataset required for this project. This project helped me practice MySQL concepts using a retail sales dataset and understand how SQL can be used to retrieve, filter, summarize and analyze business data.

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1. INTRODUCTION

This project focuses on Retail Sales Data Analysis using MySQL. The dataset contains information such as Ship mode, Segment, Country, State, City, Postal Code, Category, Sub-category, Quantity, Discount, Sales and Profit. SQL is used to retrieve useful information, perform calculations, group transactions, filter business results and demonstrate different JOIN operations.

2. PROJECT OBJECTIVES

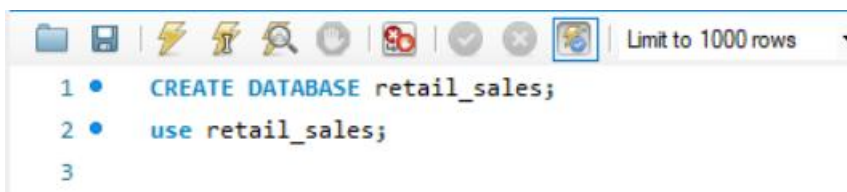
- Understand the structure of a retail sales dataset.
- Practice SELECT, WHERE, DISTINCT, ORDER BY and LIMIT.
- Use AND, OR, BETWEEN, IN and LIKE conditions.
- Apply COUNT, SUM, AVG, MIN and MAX aggregate functions.
- Use GROUP BY and HAVING for business summaries.
- Practice subqueries.
- Understand CTAS (CREATE TABLE AS SELECT).
- Demonstrate INNER JOIN, LEFT JOIN, RIGHT JOIN, CROSS JOIN and SELF JOIN.
- Present SQL output clearly using MySQL Workbench screenshots.

3. TOOLS USED

- MySQL / MySQL Workbench
- Microsoft Excel – source dataset
- Microsoft Word – project documentation

4. DATABASE SETUP

```
CREATE DATABASE retail_sales;  
USE retail_sales;
```



5. SQL QUERIES – 28 QUERIES

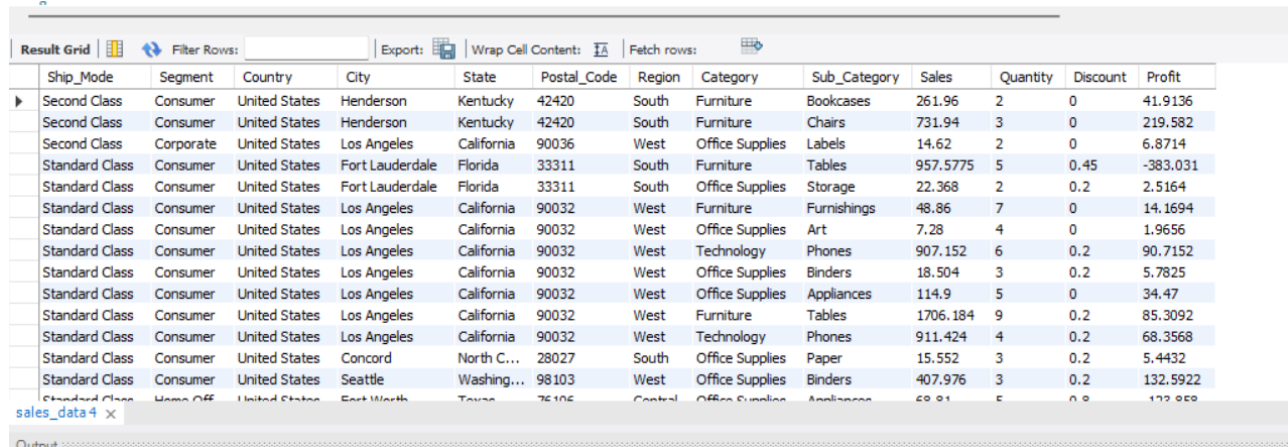
Query 1: Display all records

Objective: View all columns and rows in the sales table.

SQL Query:

```
SELECT * FROM sales_data;
```

```
4 • select * from sales_data;
5
6
7
8
```



The screenshot shows a SQL query result grid with 14 columns: Ship_Mode, Segment, Country, City, State, Postal_Code, Region, Category, Sub_Category, Sales, Quantity, Discount, and Profit. The data is displayed in a table format with alternating row colors. The first few rows are:

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.96	2	0	41.9136
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.62	2	0	6.8714

Result/Observation: Displays the complete sales dataset.

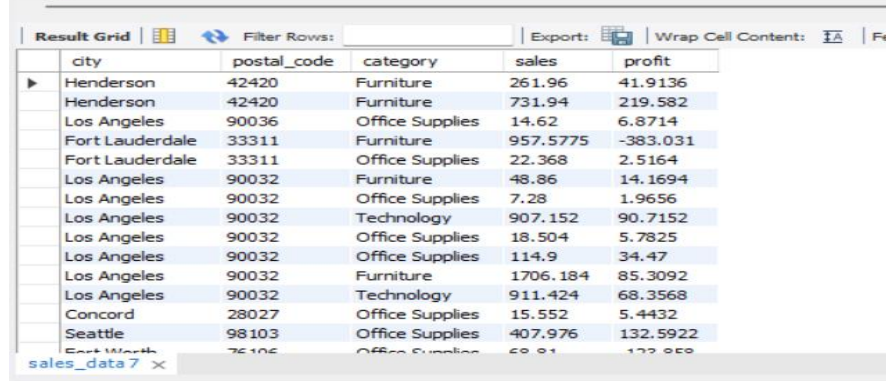
Query 2: Display selected columns

Objective: Display only important sales analysis columns.

SQL Query:

```
SELECT city, postal_code, category, sales, profit
FROM sales_data;
```

```
6 • SELECT city, postal_code, category, sales, profit
7 FROM sales_data;
```



The screenshot shows a SQL query result grid with 5 columns: city, postal_code, category, sales, and profit. The data is displayed in a table format with alternating row colors. The first few rows are:

city	postal_code	category	sales	profit
Henderson	42420	Furniture	261.96	41.9136
Henderson	42420	Furniture	731.94	219.582
Los Angeles	90036	Office Supplies	14.62	6.8714

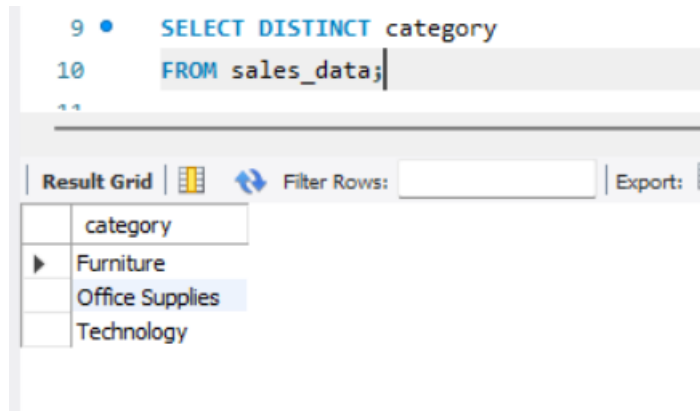
Result/Observation: Displays the selected columns.

Query 3: Find unique categories

Objective: Identify all different product categories.

SQL Query:

```
SELECT DISTINCT category FROM sales_data;
```



Result/Observation: Returns each category only once.

Query 4: Sales greater than 5000

Objective: Find orders where sales are above 5000.

SQL Query:

```
SELECT * FROM sales_data WHERE sales > 5000;
```

The screenshot shows a SQL query editor with the query `SELECT * FROM sales_data WHERE sales > 5000;` entered. Below the query, there is a 'Result Grid' tab. The results are displayed in a table with 14 columns: Ship_Mode, Segment, Country, City, State, Postal_Code, Region, Category, Sub_Category, Sales, Quantity, Discount, and Profit. The table contains 20 rows of data, all of which have a 'Sales' value greater than 5000.

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Standard Class	Consumer	United States	San Antonio	Texas	78207	Central	Technology	Machines	8159.952	8	0.4	-1359.992
Standard Class	Consumer	United States	Atlanta	Georgia	30318	South	Office Supplies	Binders	6354.95	5	0	3177.475
Same Day	Corporate	United States	Burlington	North Carolina	27217	South	Technology	Machines	7999.98	4	0.5	-3839.9904
Second Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Supplies	8187.65	5	0	327.506
First Class	Home Office	United States	New York City	New York	10024	East	Technology	Copiers	11199.968	4	0.2	3919.9888
Standard Class	Home Office	United States	Jacksonville	Florida	32216	South	Technology	Machines	22638.48	6	0.5	-1811.0784
Standard Class	Consumer	United States	Minneapolis	Minnesota	55407	Central	Office Supplies	Binders	9449.95	5	0	4630.4755
Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology	Copiers	10499.97	3	0	5039.9856
Standard Class	Corporate	United States	Lakewood	New Jersey	8701	East	Technology	Machines	9099.93	7	0	2365.9818
Standard Class	Corporate	United States	New York City	New York	10035	East	Technology	Machines	5199.96	4	0	1351.9896
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Technology	Copiers	8399.976	4	0.4	1119.9968
Second Class	Consumer	United States	Jackson	Michigan	49201	Central	Office Supplies	Binders	5443.96	4	0	2504.2216
Standard Class	Consumer	United States	New York City	New York	10024	East	Technology	Machines	6999.96	4	0	2239.9872
Standard Class	Corporate	United States	Lafayette	Indiana	47905	Central	Technology	Copiers	17499.95	5	0	8399.976
Standard Class	Home Office	United States	Providence	Rhode Island	2908	East	Technology	Copiers	5399.91	9	0	2591.9568
First Class	Consumer	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	0	6719.9808
Second Class	Consumer	United States	Arlington	Virginia	22204	South	Technology	Machines	8749.95	5	0	2799.984
Standard Class	Corporate	United States	Sacramento	California	95823	West	Office Supplies	Binders	5083.96	5	0.2	1906.485
Standard Class	Consumer	United States	Detroit	Michigan	48205	Central	Office Supplies	Binders	9892.74	13	0	4946.37

Result/Observation: Returns high-value sales transactions.

Query 5: AND condition

Objective: Find Furniture orders with sales above 3000.

SQL Query:

```
SELECT * FROM sales_data WHERE category = 'Furniture' AND sales > 3000;
```

```
15 • SELECT * FROM sales_data
16 WHERE category = 'Furniture' AND sales > 3000;
17
```

	Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
▶	Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Bookcases	3083.43	7	0.5	-1665.0522
	Second Class	Consumer	United States	Fresno	California	93727	West	Furniture	Tables	3610.848	12	0.2	135.4068
	Second Class	Corporate	United States	Detroit	Michigan	48205	Central	Furniture	Chairs	3504.9	5	0	700.98
	Standard Class	Consumer	United States	New York City	New York	10035	East	Furniture	Bookcases	4228.704	6	0.2	158.5764
	Standard Class	Consumer	United States	Buffalo	New York	14215	East	Furniture	Bookcases	4007.84	10	0.2	-50.098
	Second Class	Consumer	United States	Philadelphia	Pennsylvania	19134	East	Furniture	Chairs	4416.174	9	0.3	-630.882
	Second Class	Consumer	United States	New York City	New York	10035	East	Furniture	Chairs	3785.292	6	0.1	420.588
	Second Class	Consumer	United States	Concord	North Carolina	28027	South	Furniture	Tables	4297.644	13	0.4	-1862.3124
	Standard Class	Consumer	United States	Los Angeles	California	90045	West	Furniture	Bookcases	3406.664	8	0.15	160.3136
	Standard Class	Corporate	United States	Burlington	Vermont	5408	East	Furniture	Bookcases	4404.9	5	0	1013.127
	Standard Class	Corporate	United States	Seattle	Washington	98103	West	Furniture	Tables	3393.68	8	0	610.8624

Result/Observation: Both conditions must be true.

Query 6: OR condition

Objective: Find quantity from either Technology or Furniture.

SQL Query:

```
SELECT category, quantity FROM sales_data WHERE category = 'Technology' OR category = 'Furniture';
```

```
18 • SELECT category, quantity
19 FROM sales_data
20 WHERE category = 'Technology' OR category = 'Furniture';
```

	category	quantity
▶	Furniture	2
	Furniture	3
	Furniture	5
	Furniture	7
	Technology	6
	Furniture	9
	Technology	4
	Technology	3
	Furniture	2
	Furniture	3
	Technology	3
	Furniture	7
	Furniture	3
	Technology	7
	Furniture	5
	Furniture	3
	Furniture	3
	Technology	4
	Technology	4
	Technology	2
	Technology	3
	Technology	2
	Furniture	2
	Furniture	1
	Technology	5

sales_data 2 x

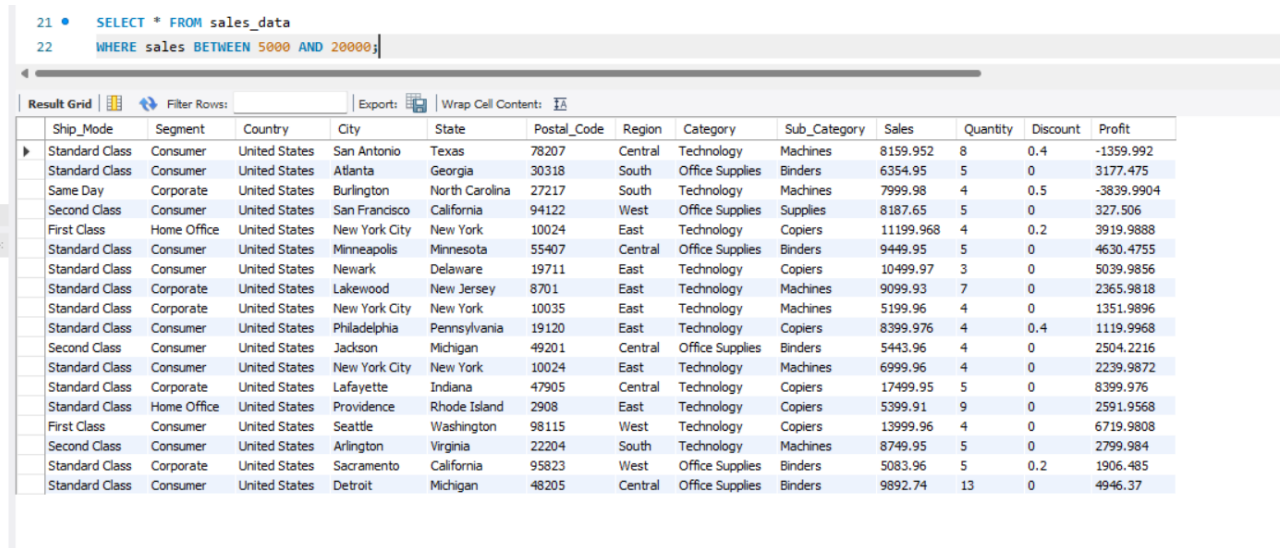
Result/Observation: Returns quantity matching either category.

Query 7: BETWEEN condition

Objective: Find sales amounts between 5000 and 20000.

SQL Query:

```
SELECT * FROM sales_data WHERE sales BETWEEN 5000 AND 20000;
```



The screenshot shows a SQL query editor with the following query:

```
21 • SELECT * FROM sales_data
22 WHERE sales BETWEEN 5000 AND 20000;
```

The results are displayed in a table grid with the following columns: Ship_Mode, Segment, Country, City, State, Postal_Code, Region, Category, Sub_Category, Sales, Quantity, Discount, Profit. The table contains 20 rows of data, all of which have sales values between 5000 and 20000.

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Standard Class	Consumer	United States	San Antonio	Texas	78207	Central	Technology	Machines	8159.952	8	0.4	-1359.992
Standard Class	Consumer	United States	Atlanta	Georgia	30318	South	Office Supplies	Binders	6354.95	5	0	3177.475
Same Day	Corporate	United States	Burlington	North Carolina	27217	South	Technology	Machines	7999.98	4	0.5	-3839.9904
Second Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Supplies	8187.65	5	0	327.506
First Class	Home Office	United States	New York City	New York	10024	East	Technology	Copiers	11199.968	4	0.2	3919.9888
Standard Class	Consumer	United States	Minneapolis	Minnesota	55407	Central	Office Supplies	Binders	9449.95	5	0	4630.4755
Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology	Copiers	10499.97	3	0	5039.9856
Standard Class	Corporate	United States	Lakewood	New Jersey	8701	East	Technology	Machines	9099.93	7	0	2365.9818
Standard Class	Corporate	United States	New York City	New York	10035	East	Technology	Machines	5199.96	4	0	1351.9896
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Technology	Copiers	8399.976	4	0.4	1119.9968
Second Class	Consumer	United States	Jackson	Michigan	49201	Central	Office Supplies	Binders	5443.96	4	0	2504.2216
Standard Class	Consumer	United States	New York City	New York	10024	East	Technology	Machines	6999.96	4	0	2239.9872
Standard Class	Corporate	United States	Lafayette	Indiana	47905	Central	Technology	Copiers	17499.95	5	0	8399.976
Standard Class	Home Office	United States	Providence	Rhode Island	2908	East	Technology	Copiers	5399.91	9	0	2591.9568
First Class	Consumer	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	0	6719.9808
Second Class	Consumer	United States	Arlington	Virginia	22204	South	Technology	Machines	8749.95	5	0	2799.984
Standard Class	Corporate	United States	Sacramento	California	95823	West	Office Supplies	Binders	5083.96	5	0.2	1906.485
Standard Class	Consumer	United States	Detroit	Michigan	48205	Central	Office Supplies	Binders	9892.74	13	0	4946.37

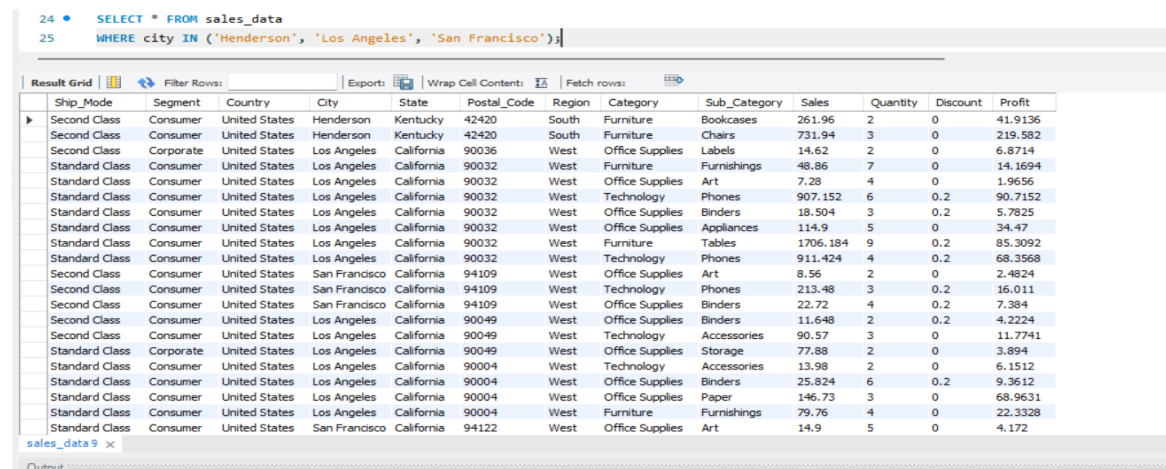
Result/Observation: Returns sales within the specified range.

Query 8: IN condition

Objective: Find details from selected cities.

SQL Query:

```
SELECT * FROM sales_data WHERE city IN ('Henderson', 'Los Angeles', 'San Francisco');
```



The screenshot shows a SQL query editor with the following query:

```
24 • SELECT * FROM sales_data
25 WHERE city IN ('Henderson', 'Los Angeles', 'San Francisco');
```

The results are displayed in a table grid with the following columns: Ship_Mode, Segment, Country, City, State, Postal_Code, Region, Category, Sub_Category, Sales, Quantity, Discount, Profit. The table contains 20 rows of data, all of which have cities in the specified list.

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.96	2	0	41.9136
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.62	2	0	6.8714
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Furnishings	48.86	7	0	14.1694
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Art	7.28	4	0	1.9656
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Binders	18.504	3	0.2	5.7825
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Appliances	114.9	5	0	34.47
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
Second Class	Consumer	United States	San Francisco	California	94109	West	Office Supplies	Art	8.56	2	0	2.4824
Second Class	Consumer	United States	San Francisco	California	94109	West	Technology	Phones	213.48	3	0.2	16.011
Second Class	Consumer	United States	San Francisco	California	94109	West	Office Supplies	Binders	22.72	4	0.2	7.384
Second Class	Consumer	United States	Los Angeles	California	90049	West	Office Supplies	Binders	11.648	2	0.2	4.2224
Second Class	Consumer	United States	Los Angeles	California	90049	West	Technology	Accessories	90.57	3	0	11.7741
Standard Class	Consumer	United States	Los Angeles	California	90049	West	Office Supplies	Storage	77.88	2	0	3.894
Standard Class	Consumer	United States	Los Angeles	California	90004	West	Technology	Accessories	13.98	2	0	6.1512
Standard Class	Consumer	United States	Los Angeles	California	90004	West	Office Supplies	Binders	25.824	6	0.2	9.3612
Standard Class	Consumer	United States	Los Angeles	California	90004	West	Office Supplies	Paper	146.73	3	0	68.9631
Standard Class	Consumer	United States	Los Angeles	California	90004	West	Furniture	Furnishings	79.76	4	0	22.3328
Standard Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Art	14.9	5	0	4.172

Result/Observation: Returns orders from the listed cities.

Query 9: LIKE operator

Objective: Find state whose name starts with 'C' and end with 'a'.

SQL Query:

```
SELECT * FROM sales_data WHERE state LIKE 'C__i%a';
```

27 • `SELECT * FROM sales_data`
28 `WHERE state LIKE 'C__i%a';`
29

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
▶	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.62	2	0	6.8714
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Furnishings	48.86	7	0	14.1694
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Art	7.28	4	0	1.9656
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Binders	18.504	3	0.2	5.7825
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Appliances	114.9	5	0	34.47
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
	Second Class	Consumer	United States	San Francisco	California	94109	West	Office Supplies	Art	8.56	2	0	2.4824
	Second Class	Consumer	United States	San Francisco	California	94109	West	Technology	Phones	213.48	3	0.2	16.011
	Second Class	Consumer	United States	San Francisco	California	94109	West	Office Supplies	Binders	22.72	4	0.2	7.384
	Second Class	Consumer	United States	Los Angeles	California	90049	West	Office Supplies	Binders	11.648	2	0.2	4.2224
	Second Class	Consumer	United States	Los Angeles	California	90049	West	Technology	Accessories	90.57	3	0	11.7741
	Standard Class	Corporate	United States	Los Angeles	California	90049	West	Office Supplies	Storage	77.88	2	0	3.894
	Standard Class	Consumer	United States	Los Angeles	California	90004	West	Technology	Accessories	13.98	2	0	6.1512
	Standard Class	Consumer	United States	Los Angeles	California	90004	West	Office Supplies	Binders	25.824	6	0.2	9.3612
	Standard Class	Consumer	United States	Los Angeles	California	90004	West	Office Supplies	Paper	146.73	3	0	68.9631
	Standard Class	Consumer	United States	Los Angeles	California	90004	West	Furniture	Furnishings	79.76	4	0	22.3328
	Standard Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Art	14.9	5	0	4.172
	Standard Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Storage	21.39	1	0	6.2031
	Standard Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Art	20.1	3	0	6.633
	Standard Class	Corporate	United States	Los Angeles	California	90036	West	Technology	Phones	73.584	2	0.2	8.2782
	Standard Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Paper	6.48	1	0	3.1104
	First Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Binders	51.312	3	0.2	17.9592

sales_data 12 x

Result/Observation: Returns state such as California

Query 10: ORDER BY ascending

Objective: Arrange transactions by sales from lowest to highest.

SQL Query:

```
SELECT * FROM sales_data ORDER BY sales ASC;
```

30 • `SELECT * FROM sales_data`
31 `ORDER BY sales ASC;`
32

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
▶	Standard Class	Consumer	United States	Houston	Texas	77095	Central	Office Supplies	Appliances	0.444	1	0.8	-1.11
	Same Day	Corporate	United States	Waco	Texas	76706	Central	Office Supplies	Binders	0.556	1	0.8	-0.9452
	Standard Class	Corporate	United States	Chicago	Illinois	60623	Central	Office Supplies	Binders	0.836	1	0.8	-1.3376
	Same Day	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Office Supplies	Binders	0.852	1	0.7	-0.5964
	Same Day	Corporate	United States	Mesquite	Texas	75150	Central	Office Supplies	Binders	0.876	1	0.8	-1.4016
	Standard Class	Corporate	United States	Huntsville	Texas	77340	Central	Office Supplies	Binders	0.898	1	0.8	-1.5715
	First Class	Consumer	United States	Houston	Texas	77041	Central	Office Supplies	Binders	0.984	2	0.8	-1.476
	First Class	Home Office	United States	San Francisco	California	94110	West	Technology	Accessories	0.99	1	0	0.4356
	Standard Class	Corporate	United States	Dallas	Texas	75217	Central	Office Supplies	Binders	1.044	1	0.8	-1.837
	Standard Class	Consumer	United States	Portland	Oregon	97206	West	Office Supplies	Binders	1.08	2	0.7	-0.792
	Standard Class	Consumer	United States	Dallas	Texas	75220	Central	Office Supplies	Binders	1.08	3	0.8	-1.728
	Standard Class	Consumer	United States	Aurora	Colorado	80013	West	Office Supplies	Binders	1.08	2	0.7	-0.792
	Same Day	Consumer	United States	Houston	Texas	77041	Central	Office Supplies	Binders	1.112	2	0.8	-1.8904
	Standard Class	Corporate	United States	Jacksonville	Florida	32216	South	Office Supplies	Binders	1.167	1	0.7	-0.8558
	Second Class	Corporate	United States	Loveland	Colorado	80538	West	Office Supplies	Binders	1.188	2	0.7	-0.99
	First Class	Corporate	United States	Philadelphia	Pennsylvania	19143	East	Office Supplies	Binders	1.188	2	0.7	-0.99
	Standard Class	Consumer	United States	Houston	Texas	77095	Central	Office Supplies	Binders	1.188	1	0.8	-1.9602
	Standard Class	Consumer	United States	Houston	Texas	77041	Central	Office Supplies	Binders	1.192	2	0.8	-2.0264
	Standard Class	Consumer	United States	Houston	Texas	77095	Central	Office Supplies	Binders	1.234	1	0.8	-1.9744
	Standard Class	Consumer	United States	Auburn	New York	13021	East	Office Supplies	Fasteners	1.24	1	0	0.5828
	First Class	Corporate	United States	Houston	Texas	77041	Central	Office Supplies	Binders	1.248	3	0.8	-1.9344
	Standard Class	Consumer	United States	Houston	Texas	77070	Central	Office Supplies	Binders	1.248	2	0.8	-1.9344
	Standard Class	Consumer	United States	San Antonio	Texas	78207	Central	Office Supplies	Binders	1.272	2	0.8	-2.1624
	Standard Class	Corporate	United States	Arlington	Texas	76017	Central	Office Supplies	Binders	1.344	4	0.8	-2.1504

sales_data 16 x

Result/Observation: Records are sorted in ascending sales order.

Query 11: ORDER BY descending

Objective: Arrange transactions by profit from highest to lowest.

SQL Query:

```
SELECT * FROM sales_data ORDER BY profit DESC;
```

33 • SELECT * FROM sales_data
34 ORDER BY profit DESC;
35

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: | Fetch rows:

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Standard Class	Corporate	United States	Lafayette	Indiana	47905	Central	Technology	Copiers	17499.95	5	0	8399.976
First Class	Consumer	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	0	6719.9808
Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology	Copiers	10499.97	3	0	5039.9856
Standard Class	Consumer	United States	Detroit	Michigan	48205	Central	Office Supplies	Binders	9892.74	13	0	4946.37
Standard Class	Consumer	United States	Minneapolis	Minnesota	55407	Central	Office Supplies	Binders	9449.95	5	0	4630.4755
First Class	Home Office	United States	New York City	New York	10024	East	Technology	Copiers	11199.968	4	0.2	3919.9888
Standard Class	Consumer	United States	Atlanta	Georgia	30318	South	Office Supplies	Binders	6354.95	5	0	3177.475
Second Class	Consumer	United States	Arlington	Virginia	22204	South	Technology	Machines	8749.95	5	0	2799.984
Standard Class	Home Office	United States	Providence	Rhode Island	2908	East	Technology	Copiers	5399.91	9	0	2591.9568
Second Class	Consumer	United States	Jackson	Michigan	49201	Central	Office Supplies	Binders	5443.96	4	0	2504.2216
Standard Class	Consumer	United States	Yonkers	New York	10701	East	Technology	Machines	4899.93	7	0	2400.9657
Standard Class	Corporate	United States	Lakewood	New Jersey	8701	East	Technology	Machines	9099.93	7	0	2365.9818
Standard Class	Corporate	United States	Springfield	Missouri	65807	Central	Technology	Copiers	4899.93	7	0	2302.9671
Standard Class	Consumer	United States	New York City	New York	10024	East	Technology	Machines	6999.96	4	0	2239.9872
Same Day	Corporate	United States	New York City	New York	10024	East	Technology	Machines	4643.8	4	0	2229.024
Standard Class	Home Office	United States	New York City	New York	10024	East	Technology	Machines	3991.98	2	0	1995.99
First Class	Home Office	United States	Midland	Michigan	48640	Central	Office Supplies	Binders	3812.97	3	0	1906.485
Standard Class	Corporate	United States	Sacramento	California	95823	West	Office Supplies	Binders	5083.96	5	0.2	1906.485
Standard Class	Home Office	United States	New York City	New York	10024	East	Technology	Machines	3404.5	5	0	1668.205
Second Class	Home Office	United States	North Las Ve...	Nevada	89031	West	Office Supplies	Binders	4535.976	3	0.2	1644.2913
First Class	Corporate	United States	Seattle	Washington	98115	West	Technology	Copiers	3149.93	7	0	1480.4671
First Class	Corporate	United States	Mobile	Alabama	36608	South	Technology	Machines	3040	8	0	1459.2
Standard Class	Consumer	United States	New York City	New York	10035	East	Office Supplies	Binders	4305.552	6	0.2	1453.1238
Second Class	Corporate	United States	Atlanta	Georgia	30318	South	Technology	Copiers	2999.95	5	0	1439.976

sales_data 17 x

Result/Observation: Records are sorted in descending profit order.

Query 12: LIMIT

Objective: Display the top 10 highest-sales transactions.

SQL Query:

```
SELECT * FROM sales_data ORDER BY sales DESC LIMIT 10;
```

36 • SELECT * FROM sales_data
37 ORDER BY sales DESC LIMIT 10;
38

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: | Fetch rows:

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Standard Class	Home Office	United States	Jacksonville	Florida	32216	South	Technology	Machines	22638.48	6	0.5	-1811.0784
Standard Class	Corporate	United States	Lafayette	Indiana	47905	Central	Technology	Copiers	17499.95	5	0	8399.976
First Class	Consumer	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	0	6719.9808
First Class	Home Office	United States	New York City	New York	10024	East	Technology	Copiers	11199.968	4	0.2	3919.9888
Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology	Copiers	10499.97	3	0	5039.9856
Standard Class	Consumer	United States	Detroit	Michigan	48205	Central	Office Supplies	Binders	9892.74	13	0	4946.37
Standard Class	Consumer	United States	Minneapolis	Minnesota	55407	Central	Office Supplies	Binders	9449.95	5	0	4630.4755
Standard Class	Corporate	United States	Lakewood	New Jersey	8701	East	Technology	Machines	9099.93	7	0	2365.9818
Second Class	Consumer	United States	Arlington	Virginia	22204	South	Technology	Machines	8749.95	5	0	2799.984
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Technology	Copiers	8399.976	4	0.4	1119.9968

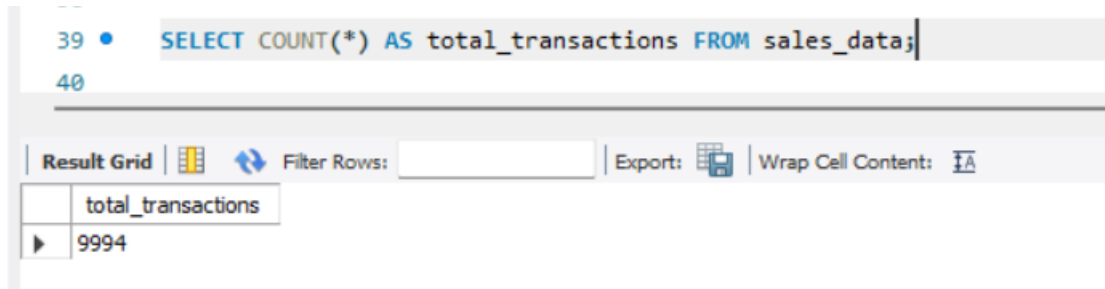
Result/Observation: Displays only the first 10 records after sorting.

Query 13: COUNT

Objective: Count the total number of sales transactions.

SQL Query:

```
SELECT COUNT(*) AS total_transactions FROM sales_data;
```



The screenshot shows a SQL query editor with the query: `SELECT COUNT(*) AS total_transactions FROM sales_data;`. Below the query, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The result grid displays a single row with the column name 'total_transactions' and the value '9994'.

total_transactions
9994

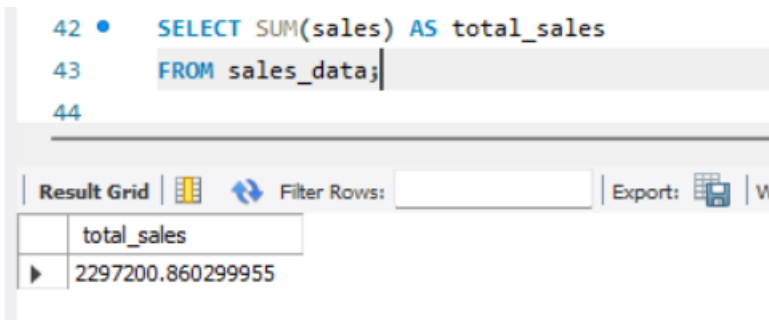
Result/Observation: Returns the total number of rows/transactions.

Query 14: SUM

Objective: Calculate total sales.

SQL Query:

```
SELECT SUM(sales) AS total_sales FROM sales_data;
```



The screenshot shows a SQL query editor with the query: `SELECT SUM(sales) AS total_sales FROM sales_data;`. Below the query, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The result grid displays a single row with the column name 'total_sales' and the value '2297200.860299955'.

total_sales
2297200.860299955

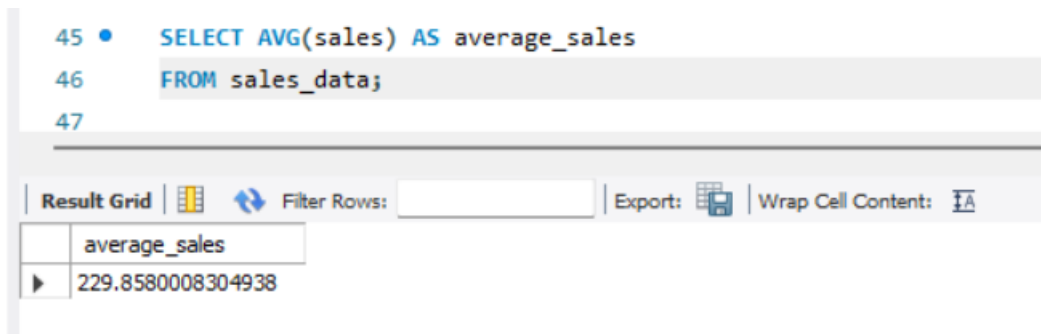
Result/Observation: Returns the total sales value.

Query 15: AVG

Objective: Calculate average sales.

SQL Query:

```
SELECT AVG(sales) AS average_sales FROM sales_data;
```



```
45 • SELECT AVG(sales) AS average_sales
46 FROM sales_data;
47
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
average_sales			
▶ 229.8580008304938			

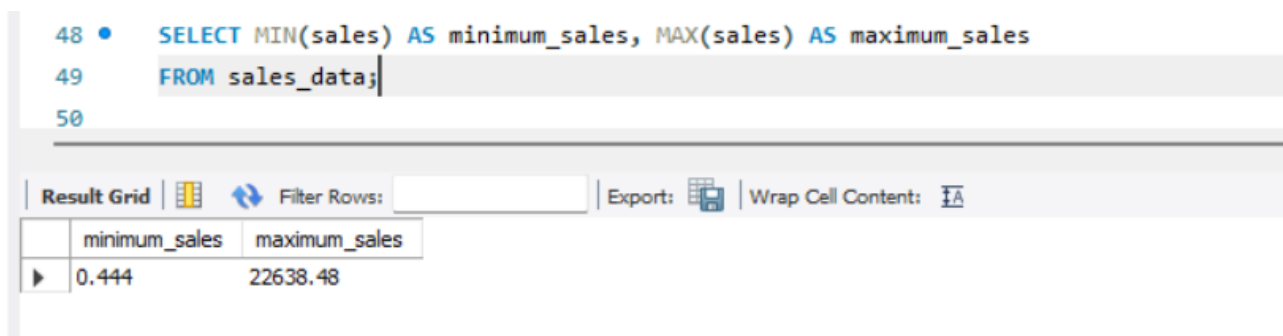
Result/Observation: Returns the average sales value.

Query 16: MIN and MAX

Objective: Find the minimum and maximum sales.

SQL Query:

```
SELECT MIN(sales) AS minimum_sales, MAX(sales) AS maximum_sales FROM sales_data;
```



```
48 • SELECT MIN(sales) AS minimum_sales, MAX(sales) AS maximum_sales
49 FROM sales_data;
50
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
minimum_sales			
▶ 0.444			
maximum_sales			
▶ 22638.48			

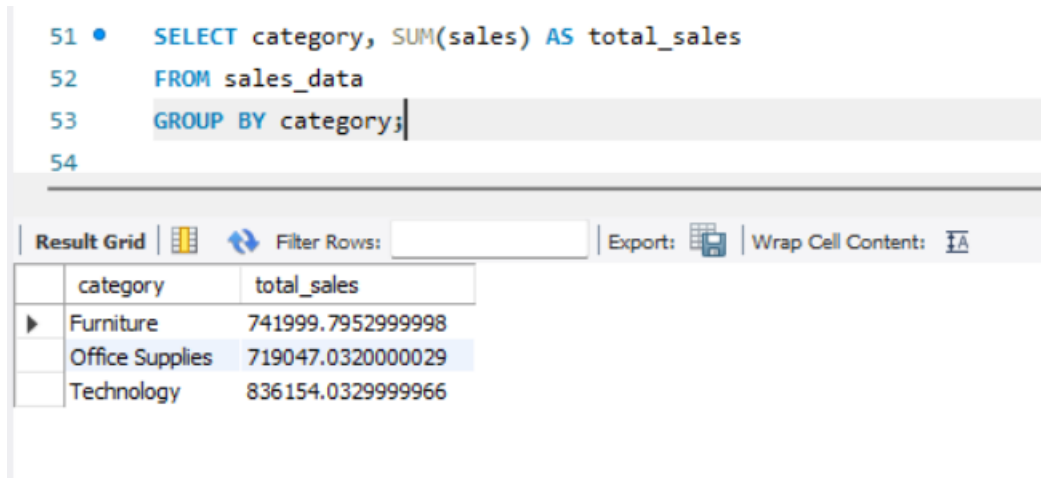
Result/Observation: Returns the lowest and highest sales values.

Query 17: GROUP BY category

Objective: Calculate total sales for each category.

SQL Query:

```
SELECT category, SUM(sales) AS total_sales FROM sales_data GROUP BY category;
```



The screenshot shows a SQL query editor with the following query:

```
51 • SELECT category, SUM(sales) AS total_sales
52 FROM sales_data
53 GROUP BY category;
54
```

Below the query editor is a toolbar with options: Result Grid, Filter Rows, Export, and Wrap Cell Content. The result grid displays the following data:

category	total_sales
Furniture	741999.7952999998
Office Supplies	719047.0320000029
Technology	836154.0329999966

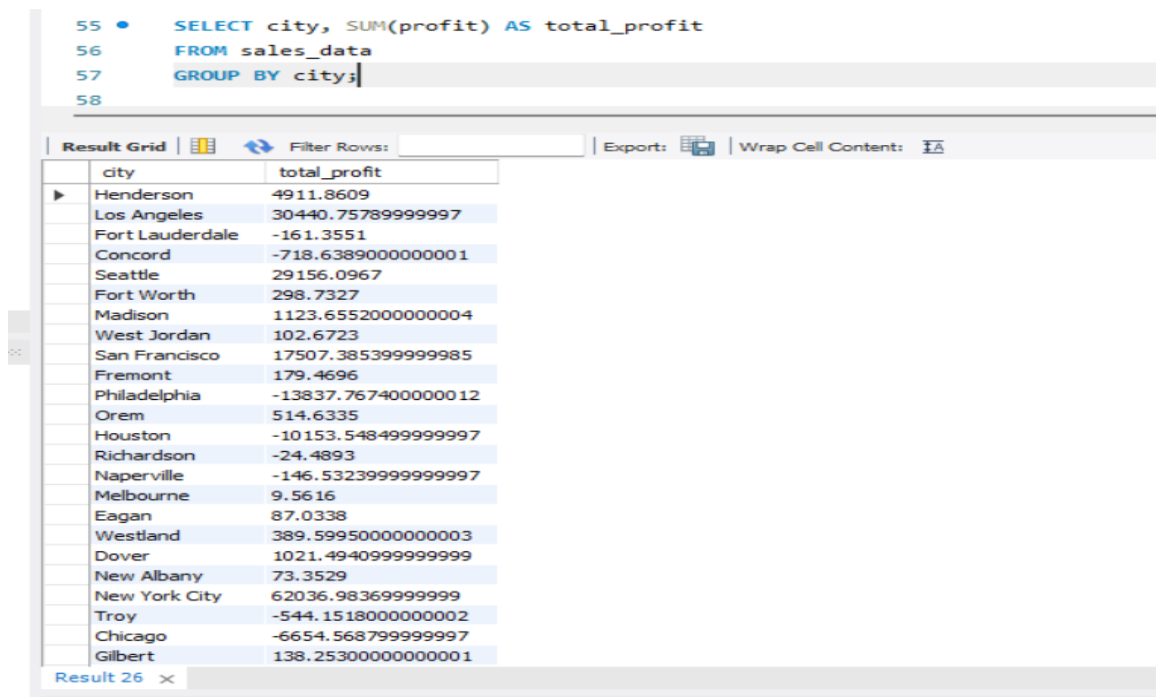
Result/Observation: Returns category-wise sales totals.

Query 18: GROUP BY city

Objective: Calculate total profit for each city.

SQL Query:

```
SELECT city, SUM(profit) AS total_profit FROM sales_data GROUP BY city;
```



The screenshot shows a SQL query editor with the following query:

```
55 • SELECT city, SUM(profit) AS total_profit
56 FROM sales_data
57 GROUP BY city;
58
```

Below the query editor is a toolbar with options: Result Grid, Filter Rows, Export, and Wrap Cell Content. The result grid displays the following data:

city	total_profit
Henderson	4911.8609
Los Angeles	30440.75789999997
Fort Lauderdale	-161.3551
Concord	-718.6389000000001
Seattle	29156.0967
Fort Worth	298.7327
Madison	1123.6552000000004
West Jordan	102.6723
San Francisco	17507.385399999985
Fremont	179.4696
Philadelphia	-13837.767400000012
Orem	514.6335
Houston	-10153.548499999997
Richardson	-24.4893
Naperville	-146.53239999999997
Melbourne	9.5616
Eagan	87.0338
Westland	389.59950000000003
Dover	1021.4940999999999
New Albany	73.3529
New York City	62036.98369999999
Troy	-544.1518000000002
Chicago	-6654.568799999997
Gilbert	138.25300000000001

Result/Observation: Returns city-wise profit totals.

Query 19: HAVING

Objective: Find categories whose total sales exceed 100000.

SQL Query:

```
SELECT category, SUM(sales) AS total_sales FROM sales_data GROUP BY category HAVING SUM(sales) > 100000;
```

The screenshot shows a SQL query editor with the following code:

```
59 • SELECT category, SUM(sales) AS total_sales
60 FROM sales_data
61 GROUP BY category
62 HAVING SUM(sales) > 100000;
63
```

Below the editor is a 'Result Grid' with the following data:

category	total_sales
Furniture	741999.7952999998
Office Supplies	719047.0320000029
Technology	836154.0329999966

Result/Observation: HAVING filters grouped results.

Query 20: Subquery – above average

Objective: Find transactions with sales greater than the overall average sales.

SQL Query:

```
SELECT * FROM sales_data WHERE sales > (SELECT AVG(sales) FROM sales_data);
```

The screenshot shows a SQL query editor with the following code:

```
72 • SELECT * FROM sales_data
73 WHERE sales > (SELECT AVG(sales)
74 FROM sales_data);
```

Below the editor is a 'Result Grid' with the following data:

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.96	2	0	41.9136
Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.031
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
Standard Class	Consumer	United States	Seattle	Washington	98103	West	Office Supplies	Binders	407.976	3	0.2	132.5922
Standard Class	Consumer	United States	Madison	Wisconsin	53711	Central	Office Supplies	Storage	665.88	6	0	13.3176
Standard Class	Consumer	United States	Orem	Utah	84057	West	Furniture	Tables	1044.63	3	0	240.2649
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Bookcases	3083.43	7	0.5	-1665.0522
First Class	Corporate	United States	Richardson	Texas	75080	Central	Technology	Phones	1097.544	7	0.2	123.4737
Standard Class	Home Of...	United States	Houston	Texas	77041	Central	Furniture	Bookcases	532.3992	3	0.32	-46.9764
Standard Class	Home Of...	United States	Houston	Texas	77041	Central	Technology	Phones	371.168	4	0.2	41.7564
Standard Class	Corporate	United States	New York City	New York	10024	East	Technology	Phones	1029.95	5	0	298.6855
First Class	Consumer	United States	Troy	New York	12180	East	Furniture	Chairs	319.41	5	0.1	7.098
Standard Class	Corporate	United States	Gilbert	Arizona	85234	West	Office Supplies	Art	1113.024	8	0.2	111.3024
Standard Class	Consumer	United States	Memphis	Tennessee	38109	South	Furniture	Chairs	831.936	8	0.2	-114.3912
First Class	Home Of...	United States	Chicago	Illinois	60623	Central	Office Supplies	Storage	230.376	3	0.2	-48.9549
Standard Class	Consumer	United States	Columbia	South Carolina	29203	South	Furniture	Chairs	301.96	2	0	33.2156
Standard Class	Consumer	United States	Avon	Colorado	80012	West	Technology	Accessories	738.806	6	0.2	-36.8768

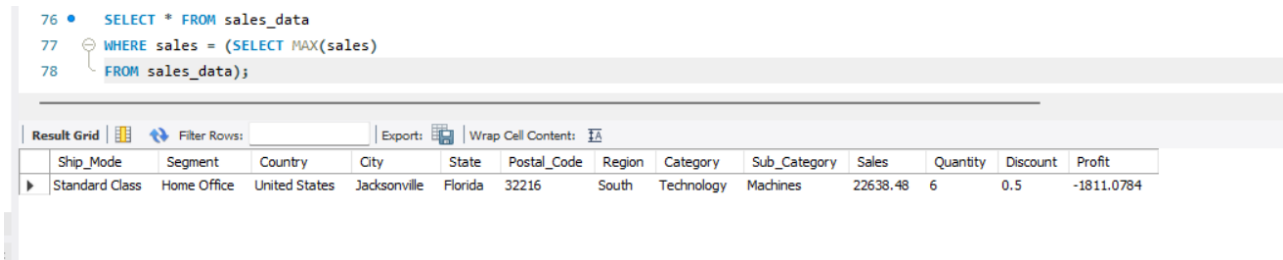
Result/Observation: Returns above-average sales transactions.

Query 21: Subquery – maximum sales

Objective: Find the transaction(s) having the maximum sales value.

SQL Query:

```
SELECT * FROM sales_data WHERE sales = (SELECT MAX(sales) FROM sales_data);
```



76 • SELECT * FROM sales_data
77 WHERE sales = (SELECT MAX(sales)
78 FROM sales_data);

Ship_Mode	Segment	Country	City	State	Postal_Code	Region	Category	Sub_Category	Sales	Quantity	Discount	Profit
Standard Class	Home Office	United States	Jacksonville	Florida	32216	South	Technology	Machines	22638.48	6	0.5	-1811.0784

Result/Observation: Returns the highest-sales transaction(s).

Query 22: CTAS – product table

Objective: Create a new table containing unique sub_category and categories.

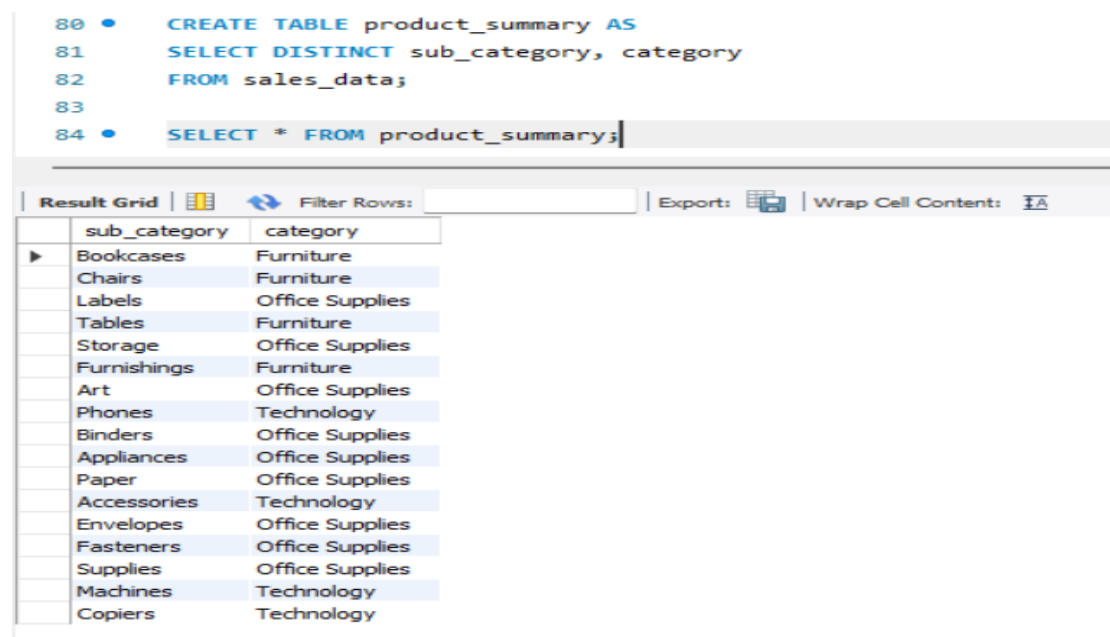
SQL Query:

```
CREATE TABLE product_summary AS  
SELECT DISTINCT sub_category, category  
FROM sales_data;
```

Result/Observation: CTAS creates a new table and copies the result rows as well as the selected columns.

Verification Query:

```
SELECT * FROM product_summary;
```



80 • CREATE TABLE product_summary AS
81 SELECT DISTINCT sub_category, category
82 FROM sales_data;
83
84 • SELECT * FROM product_summary;

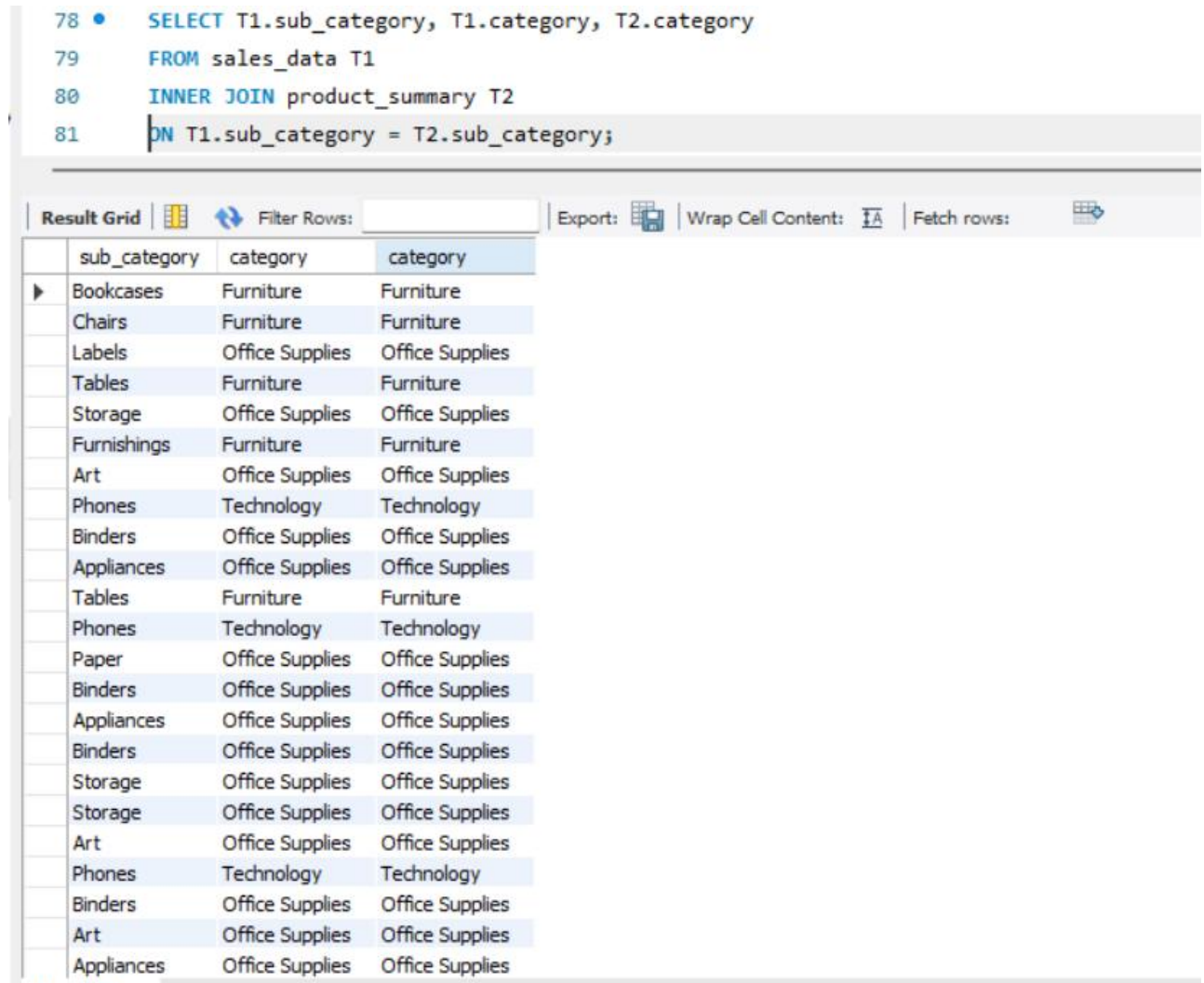
sub_category	category
Bookcases	Furniture
Chairs	Furniture
Labels	Office Supplies
Tables	Furniture
Storage	Office Supplies
Furnishings	Furniture
Art	Office Supplies
Phones	Technology
Binders	Office Supplies
Appliances	Office Supplies
Paper	Office Supplies
Accessories	Technology
Envelopes	Office Supplies
Fasteners	Office Supplies
Supplies	Office Supplies
Machines	Technology
Copiers	Technology

Query 23: INNER JOIN

Objective: Join sales_data with product_summary using sub_category

SQL Query:

```
SELECT T1.sub_category, T1.category, T2.category
FROM sales_data T1 INNER JOIN product_summary T2
ON T1.sub_category = T2.sub_category;
```



```
78 • SELECT T1.sub_category, T1.category, T2.category
79 FROM sales_data T1
80 INNER JOIN product_summary T2
81 ON T1.sub_category = T2.sub_category;
```

	sub_category	category	category
▶	Bookcases	Furniture	Furniture
	Chairs	Furniture	Furniture
	Labels	Office Supplies	Office Supplies
	Tables	Furniture	Furniture
	Storage	Office Supplies	Office Supplies
	Furnishings	Furniture	Furniture
	Art	Office Supplies	Office Supplies
	Phones	Technology	Technology
	Binders	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies
	Tables	Furniture	Furniture
	Phones	Technology	Technology
	Paper	Office Supplies	Office Supplies
	Binders	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies
	Binders	Office Supplies	Office Supplies
	Storage	Office Supplies	Office Supplies
	Storage	Office Supplies	Office Supplies
	Art	Office Supplies	Office Supplies
	Phones	Technology	Technology
	Binders	Office Supplies	Office Supplies
	Art	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies

Result/Observation: Returns only matching sub_category records from both tables.

Query 24: LEFT JOIN

Objective: Display all sales records and matching product-summary records.

SQL Query:

```
SELECT T1.sub_category, T1.category, T2.category
FROM sales_data T1 LEFT OUTER JOIN product_summary T2
ON T1.sub_category = T2.sub_category;
```

83 • SELECT T1.sub_category, T1.category, T2.category
84 FROM sales_data T1
85 LEFT OUTER JOIN product_summary T2
86 ON T1.sub_category = T2.sub_category;

Result Grid

	sub_category	category	category
►	Bookcases	Furniture	Furniture
	Chairs	Furniture	Furniture
	Labels	Office Supplies	Office Supplies
	Tables	Furniture	Furniture
	Storage	Office Supplies	Office Supplies
	Furnishings	Furniture	Furniture
	Art	Office Supplies	Office Supplies
	Phones	Technology	Technology
	Binders	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies
	Tables	Furniture	Furniture
	Phones	Technology	Technology
	Paper	Office Supplies	Office Supplies
	Binders	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies
	Binders	Office Supplies	Office Supplies
	Storage	Office Supplies	Office Supplies
	Storage	Office Supplies	Office Supplies
	Art	Office Supplies	Office Supplies
	Phones	Technology	Technology
	Binders	Office Supplies	Office Supplies
	Art	Office Supplies	Office Supplies
	Appliances	Office Supplies	Office Supplies

Result 56

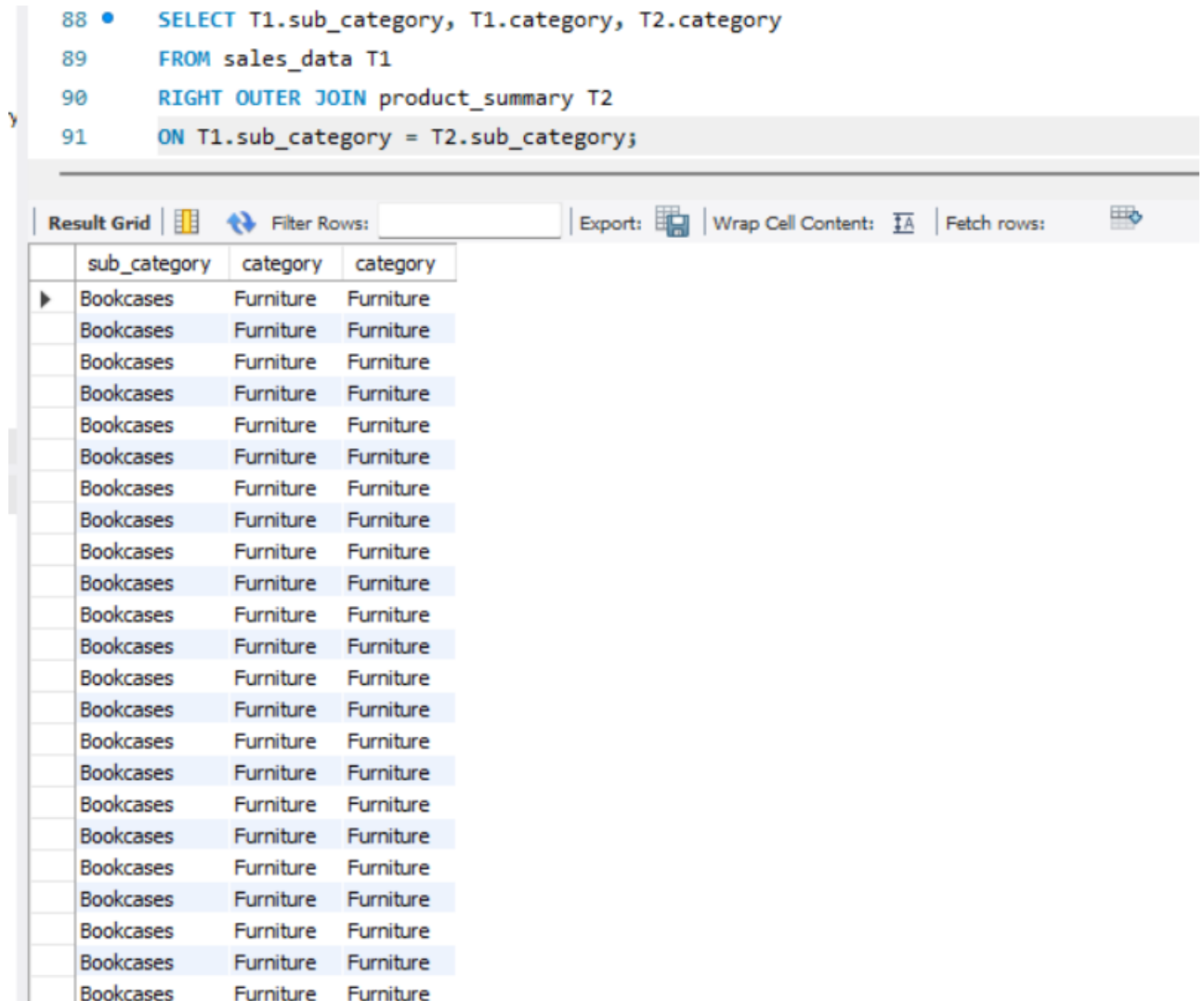
Result/Observation: All sales_data rows are retained; matching rows are taken from product_summary.

Query 25: RIGHT JOIN

Objective: Display all product-summary records and matching sales records.

SQL Query:

```
SELECT T1.sub_category, T1.category, T2.category
FROM sales_data T1 RIGHT OUTER JOIN product_summary T2
ON T1.sub_category = T2.sub_category;
```



```
88 • SELECT T1.sub_category, T1.category, T2.category
89 FROM sales_data T1
90 RIGHT OUTER JOIN product_summary T2
91 ON T1.sub_category = T2.sub_category;
```

	sub_category	category	category
▶	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
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	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
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	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture
	Bookcases	Furniture	Furniture

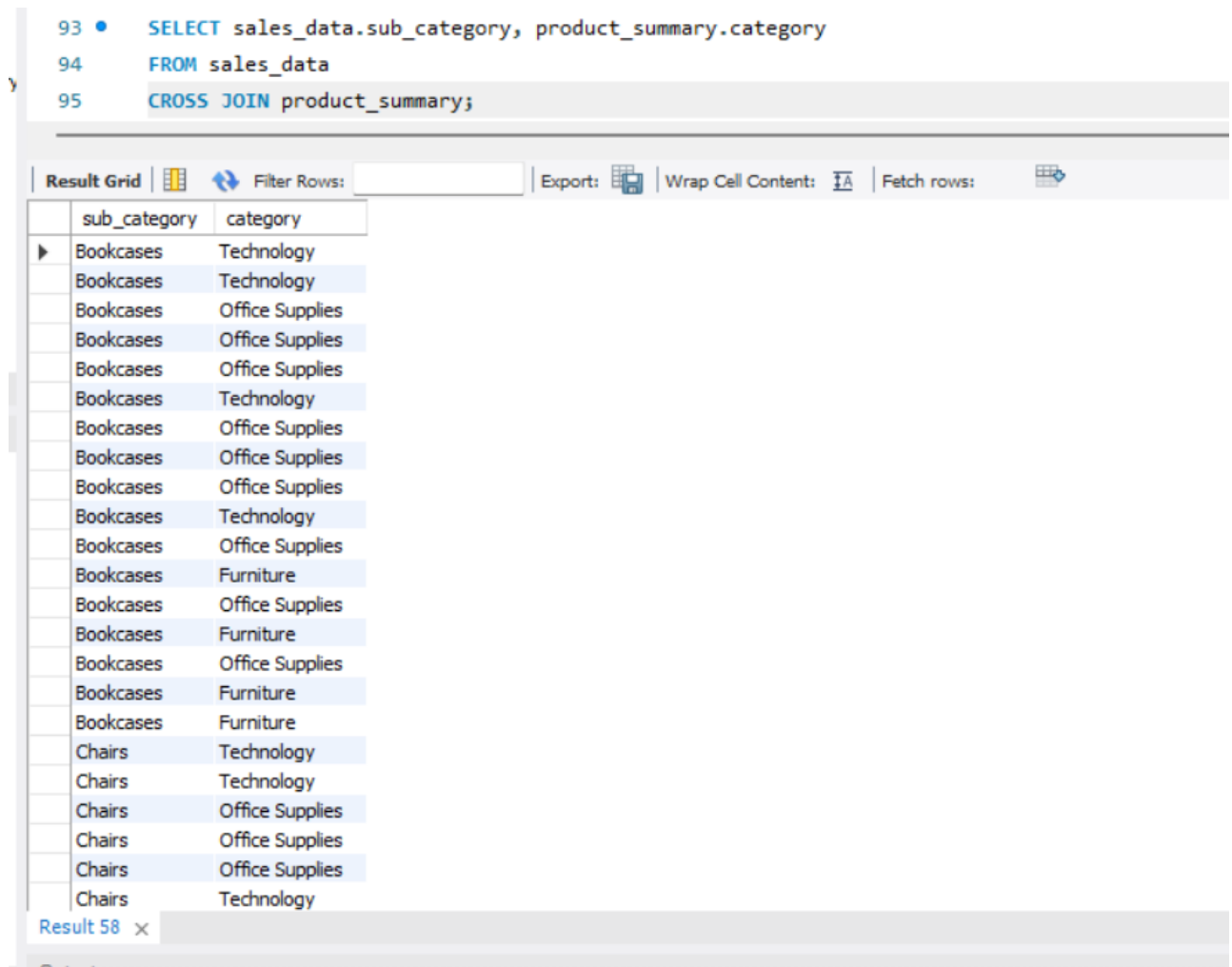
Result/Observation: All product_summary rows are retained.

Query 26: CROSS JOIN

Objective: Create combinations of sub_category and category.

SQL Query:

```
SELECT sales_data.sub_category, product_summary.category  
FROM sales_data CROSS JOIN product_summary;
```



The screenshot shows a SQL query editor with the following query:

```
93 • SELECT sales_data.sub_category, product_summary.category  
94 FROM sales_data  
95 CROSS JOIN product_summary;
```

Below the query editor is a "Result Grid" tab. The grid displays the results of the query, showing all possible combinations of sub_category and category. The grid has two columns: "sub_category" and "category".

sub_category	category
Bookcases	Technology
Bookcases	Technology
Bookcases	Office Supplies
Bookcases	Office Supplies
Bookcases	Office Supplies
Bookcases	Technology
Bookcases	Office Supplies
Bookcases	Office Supplies
Bookcases	Office Supplies
Bookcases	Office Supplies
Bookcases	Technology
Bookcases	Office Supplies
Bookcases	Furniture
Bookcases	Office Supplies
Bookcases	Furniture
Bookcases	Office Supplies
Bookcases	Furniture
Bookcases	Furniture
Chairs	Technology
Chairs	Technology
Chairs	Office Supplies
Chairs	Office Supplies
Chairs	Office Supplies
Chairs	Technology

The result grid shows 24 rows, which is the product of 8 rows in the "sub_category" table and 3 rows in the "category" table. The "sub_category" table has 8 rows: 6 "Bookcases", 1 "Chairs", and 1 "Furniture". The "category" table has 3 rows: "Technology", "Office Supplies", and "Furniture".

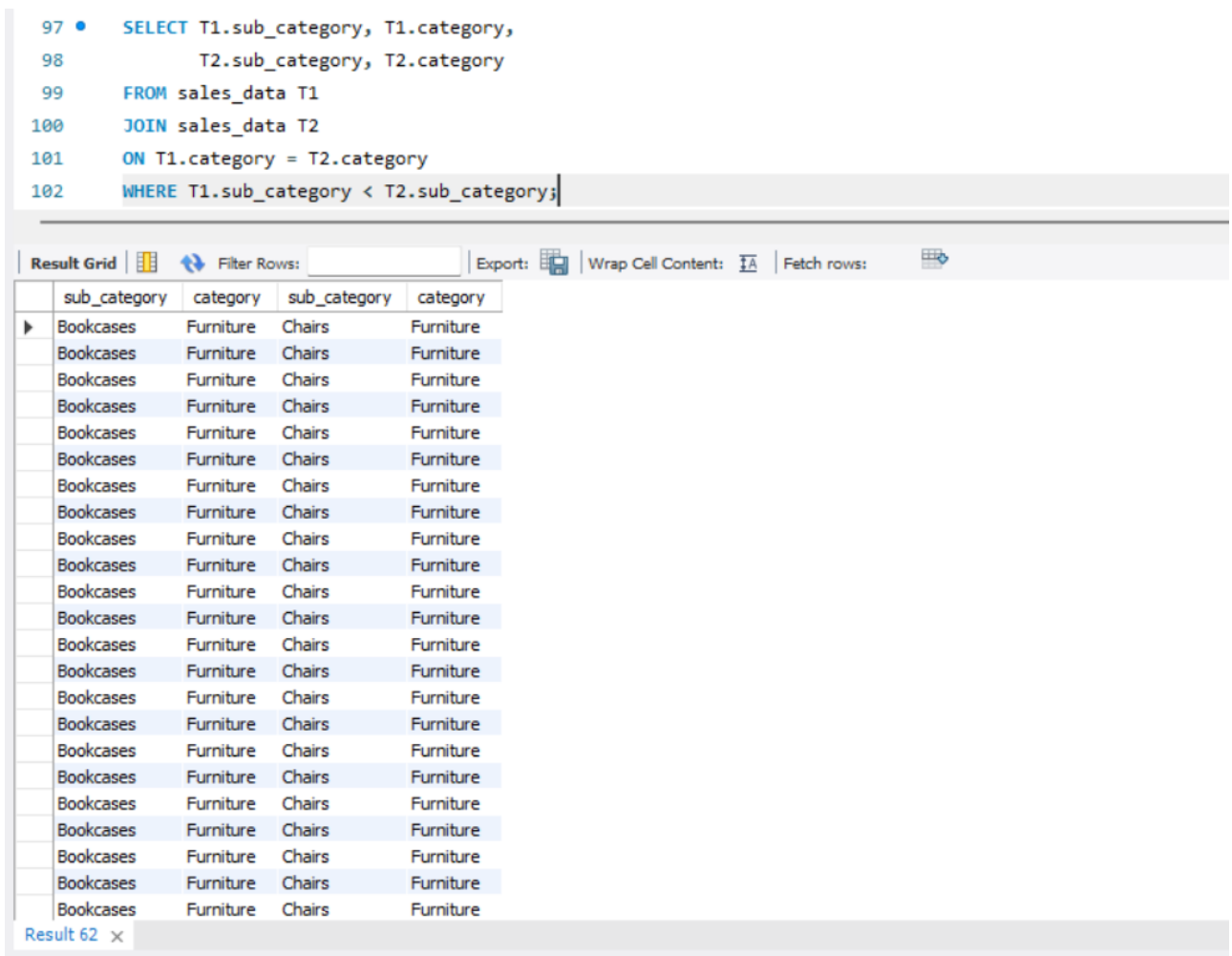
Result/Observation: It produces all possible combinations of rows from both tables.

Query 27: SELF JOIN

Objective: To join a table with itself and compare records within the same table.

SQL Query:

```
SELECT T1.sub_category, T1.category,  
       T2.sub_category, T2.category  
FROM sales_data T1 JOIN sales_data T2  
ON T1.category = T2.category  
WHERE T1.sub_category < T2.sub_category;
```



```
97 • SELECT T1.sub_category, T1.category,  
98       T2.sub_category, T2.category  
99 FROM sales_data T1  
100 JOIN sales_data T2  
101 ON T1.category = T2.category  
102 WHERE T1.sub_category < T2.sub_category;
```

	sub_category	category	sub_category	category
▶	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture
	Bookcases	Furniture	Chairs	Furniture

Result 62 x

Result/Observation: It displays pairs of different sub-categories belonging to the same category.

Query 28: CTAS – category summary

Objective: Create a summary table containing category-wise sales and profit.

SQL Query:

```
CREATE TABLE category_summary AS
SELECT category,
       SUM(sales) AS total_sales,
       SUM(profit) AS total_profit,
       AVG(sales) AS average_sales
FROM sales_data
GROUP BY category;
```

```
104 • CREATE TABLE category_summary AS
105     SELECT category,
106           SUM(sales) AS total_sales,
107           SUM(profit) AS total_profit,
108           AVG(sales) AS average_sales
109     FROM sales_data
110     GROUP BY category;
111
112
```

Result/Observation: Creates category_summary with grouped results. Verify it with SELECT * FROM category_summary;

Verification Query:

```
SELECT * FROM category_summary;
```

```
112 • SELECT * FROM category_summary;
```

	category	total_sales	total_profit	average_sales
►	Furniture	741999.7952999998	18451.2728	349.83488698727007
	Office Supplies	719047.0320000029	122490.80080000011	119.32410089611732
	Technology	836154.0329999966	145454.9480999999	452.70927612344155

6. CONCLUSION

This project provided practical experience in MySQL using a retail sales dataset. The 28 queries cover fundamental SQL operations as well as aggregate functions, grouping, HAVING, subqueries, CTAS and multiple JOIN types. The project also demonstrates how SQL can be used to turn raw transaction data into useful business information.